

Probabilistic-Bits Based on Ferroelectric Field-Effect Transistors for Probabilistic Computing

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Abstract—A probabilistic bit (p-bit) is the fundamental building block in the circuit of probabilistic computing (PC), and it produces a random binary bitstream with tunable probability. Utilizing the randomness induced by thermal noise-induced lattice vibration in the ferroelectric (FE) material, we propose the p-bits based on stochastic ferroelectric FET (FeFET). The domain dynamic is revealed to play crucial roles in FE p-bits' stochasticity, as the domain coupling suppresses the dipole fluctuation. The proposed FE p-bits possess the advantages of both extremely low hardware cost and scalability for p-bit circuitry, rendering it a promising candidate for PC.

Index Terms—FeFET, probabilistic-bit, probabilistic computing, dynamic model, multi-domain system.

I. INTRODUCTION

THE probabilistic computing (PC) [1], [2], [3], [4] is considered as an efficient solution to many hard computational problems, such as the combinatorial optimization problem (COP) [5], integer factorization (IF) [1], Bayesian inference [6], quantum simulation [7], [8] and Boltzmann-machine based machine learning [9], etc. Compared with the deterministic “bits” in traditional binary circuits, PC makes use of unstable units called the probabilistic bits (p-bits) [1] to serve as the building blocks, which could generate a continuous random digital bitstream with tunable probability. The p-bits could be realized in fully CMOS modules, such as microcontrollers [10], [11] and field programmable gate arrays (FPGA) [12]. These designs utilize the pseudo-random number generator to produce the adjustable probabilistic binary bitstream, and these designs have high transistor count (for example, 1194 for FPGA-based p-bit [12]) and energy consumption. Some different approaches to p-bit designs are to rely on various physical phenomena to induce stochasticity [1], [13], [14]. One such example is the magnetic tunnel junction (MTJ) based p-bit, which utilizes the stochastic magnetoresistance induced by thermal noise [1], [13]. The hardware cost is thus

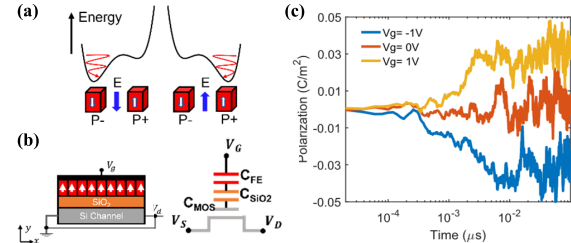


Fig. 1. (a) The mechanism of thermal noise-induced polarization fluctuation in the FE system with the free-energy profiles. The chiral red arrows represent the fluctuating polarization around the spontaneous polarization. (b) The sketch of FeFET and the corresponding capacitance model. The red color area depicts the multidomain HZO layer. (c) Polarization fluctuations under different gate biases.

significantly reduced, as each p-bit consists of a single MTJ with a resistor/transistor while the optimization of its power consumption requires further study.

Similar to the stochasticity in the ferromagnetic system, the dynamic behaviours of the ferroelectric (FE) material are random as well, enabling it as a prospective candidate for p-bits design. One of the stochastic mechanisms in the FE system stems from the thermal noise-induced lattice vibration, leading to constant dipole fluctuations [15], [16]. It plays a crucial role in the FE dynamic properties, such as the dipole reversal switching [17], [18], [19], [20], [21], [22], [23], dielectric response [24], and domain motions [18], [23]. The induced dipole fluctuations could be characterized as random Brownian motions [18] with an average fluctuating position, which could be adjusted by the external electric field (Fig. 1a). Through the FE-based devices, such as the three-terminal FeFET, the fluctuations of the dipoles could generate random current signals with adjustable range. With a designated voltage or current threshold, a mechanism of digitized random bitstream with controllable probability could be established.

In this work, we propose the stochastic p-bit based on a few-domain $\text{Hf}_x\text{Zr}_{1-x}\text{O}_2$ (HZO) FeFET (Fig. 1b), utilizing the thermal noise-induced dipole fluctuation. The crucial role of domain dynamics in the FE stochasticity is revealed in the phase-field model. The proposed FE-based p-bit is CMOS-compatible and found to possess both advantages of extremely low hardware cost and scalability, rendering it a promising candidate in large-scale PC circuits.

II. MODELLING METHODS

The thermal noise-induced dipole fluctuations in the FE system could be characterized through a statistical approach [18], [19], [20], [21] within the time-dependent Landau-Ginzburg

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TABLE I
LIST OF PARAMETERS

Parameters	Value
$\mu(\Omega \cdot \text{m})$	1.0×10^1
$\alpha_1 (\text{m}/\text{F})$	-4.0×10^8
$\alpha_{11} (\text{m}^5 \text{C}^{-2}/\text{F})$	3.7×10^9
$\alpha_{111} (\text{m}^9 \text{C}^{-4}/\text{F})$	1.1×10^9
$G_0 (\text{m}^3/\text{F})$	dynamic
$\lambda_1, \lambda_2 (\text{\AA})$	0.05, 2.5
$T (\text{K})$	300, or dynamic
$\Delta t (\text{ns})$	0.02
$V_{\text{char}} (\text{m}^3)$	1.38×10^{-29}

equations (TDLG) framework. By considering the symmetry and orientation of the HZO structure, we assume only the out-of-plane component p_y is non-zero, while the in-plane components' dynamics are ignored. The HZO is also assumed stress-free, and thus the elastic strain energy could be ignored. The overall TDLG is thus expressed as:

$$\mu \frac{\partial p(x, y, t)}{\partial t} = -\frac{\delta W}{\delta p(x, y, t)} + \nabla \cdot \frac{\delta W}{\delta \nabla p} + \mu \eta \quad (1)$$

where μ represents the viscosity coefficient, W is the enthalpy energy, and η reflects the impact of thermal noise. The enthalpy energy W could be expressed as:

$$W(\varepsilon, e, p_y, \nabla p_y) = \Psi_{\text{pol}} + \Psi_{\text{inter}} + \Psi_{\text{el}} - E_{\text{fe}} \cdot p_y - E_{\text{dep}} \cdot p_y \quad (2)$$

where Ψ_{pol} is the polarization potential energy as: $\Psi_{\text{pol}}(p_y) = \alpha_1(T)p_y(t)^2 + \alpha_{11}p_y(t)^4 + \alpha_{111}p_y(t)^6$, and $\alpha_1, \alpha_{11}, \alpha_{111}$ are the Landau coefficients [25]; Ψ_{inter} is the domain coupling energy, defined as: $\Psi_{\text{inter}}(\nabla p_y) = \frac{G_0}{2} |\nabla p_y|^2$, where G_0 is the domain coupling strength; Ψ_{el} is the vacuum energy, expressed as: $\Psi_{\text{el}} = -\frac{\varepsilon_0}{2} E_{\text{fe}}^2$, where E_{fe} is the electric field across the FE layer; E_{dep} is the depolarization field, approximated as: $E_{\text{dep}} = \frac{\lambda_1 + \lambda_2}{\hbar \varepsilon_0 + (\lambda_1 + \lambda_2) \varepsilon_b} p_y$ [26], [27], characterizing a TiN/HZO/SiO₂ gate stack with λ_1 and λ_2 are the normalized effective screening length of TiN and SiO₂ [18], [28]. Comparing to the previous work [29], the inclusion of depolarization provides a more comprehensive modelling scheme of multi-domain FE system. For η , its correlation function [18], [19] could be expressed as:

$$\langle \eta(r, t), \eta(r', t') \rangle_{t, \Omega} = \frac{2kT}{\mu} \delta(t - t') \delta(r - r') \quad (3)$$

The overall distribution of thermal noise is assumed to follow the Gaussian distribution $G_{(0,1)}$ as: $\eta = \sqrt{\frac{2k_B T}{\mu V_{\text{char}} \Delta t}} \times G_{(0,1)}$, with the mean and deviations set as 0 and 1. The V_{char} [25] and Δt represent the unit volume and time step in TDLG, and their values are shown in Table I. The fluctuation time of the polarization of the modelled FE system is at the scale of ns (Fig. 1c).

To model the dynamic behaviours of the stochastic FeFET, the quasi-static condition is assumed in each time step (~ 0.02 ns), as the response time of electron carriers is much faster than the deformation of the FE structure [30], [31]. The capacitance model is adopted to evaluate the electrostatics of

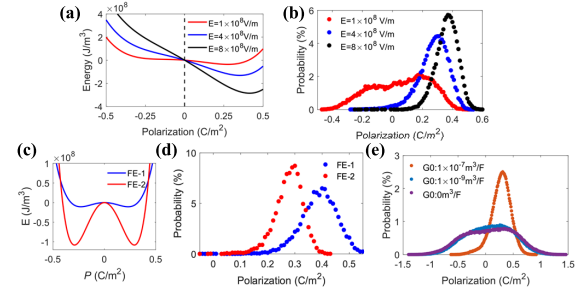


Fig. 2. (a) and (b) Free energy profiles under different electric fields and the corresponding polarization distribution. (c) Free energy profiles of FE-1 and FE-2. (d) Polarization distribution of FE-1 and FE-2 in a single domain. (e) Overall polarization distribution under different domain interaction strengths.

TiN/HZO/SiO₂/Si structures (Fig. 1b). It could be expressed as:

$$Q_{\text{FE}} = Q_{\text{SiO}_2} = Q_{\text{MOS}} \quad (4)$$

where the Q_{FE} , Q_{SiO_2} , and Q_{MOS} are the charge density of HZO layer (6 nm thick), SiO₂ layer (2 nm thick) and the MOS structure. Under the different applied voltage, the fluctuation range of the FE layer's polarization fluctuation is shifted accordingly (Fig. 1c). For the carrier transport simulations, the combined schemes of Top-of-Barrier (ToB) [32] and Wentzel–Kramers–Brillouin (WKB) [33] schemes are implemented to model the ballistic and tunnelling current in FeFET with a 10-nm long channel.

III. RESULTS

A. Thermal Noise-Induced Stochasticity in FE

To understand the mechanism of stochasticity in the HZO layer, we evaluate the impact of several key parameters within the TDLG framework. The thermal noise allows the FE system to move from the spontaneous polarization (SP). In Fig. 2a, the increasing external electric field reduces the energy of the SP and enlarges the energy differences between the opposite polarizations. The corresponding fluctuation range in Fig. 2b is restricted around the SP under the increasing applied field, and the probability of the polarization turned to the opposite direction is suppressed. The distribution profile coincides with the thermodynamic nature of the FE stochasticity.

The domain dynamics also play a crucial role in the FE system and the stochasticity of a two-domain FE system (FE-1 and FE-2) is evaluated to illustrate its impact. Each domain has a different set of Landau coefficients, and the free energy profile is shown in Fig. 2c. The domain-to-domain variations of these parameters stem from defects, doping, and other factors in the FE system [34]. Therefore, each domain has different polarization fluctuation (Fig. 2d). By adjusting the domain coupling strength (G_0), the overall fluctuation range of the two-domain system is significantly narrowed due to the formation of the domain wall (Fig. 2e). The formation of domain wall alters the original SP positions at the interface between the two domains, which effectively narrows the overall fluctuation range.

B. The Stochasticity in Multidomain FeFET

A three-terminal FeFET is implemented to convert the constantly fluctuating polarization into current signals, resulting from the fluctuating capacitance of the FE layer. To approach the multi-domain FE system, the Landau coefficients and

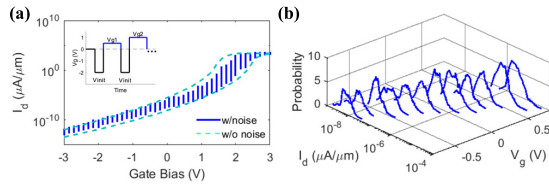


Fig. 3. (a) The I - V characteristics of the FeFET. The blue lines and cyan dash lines represent the FeFET w/ and w/o the thermal noise. Inset: the configuration of gate bias V_g . Before each gate bias, the system is initialized through V_{init} to enforce each domain starting from the same polarization. Then, different gate bias (V_{g1} , V_{g2} , ...) are applied to the system and remain constant within the observation time of 10 μ s. Approximately 10^4 time steps have been sampled in each gate bias. (b) Zoom-in view of the current distribution.

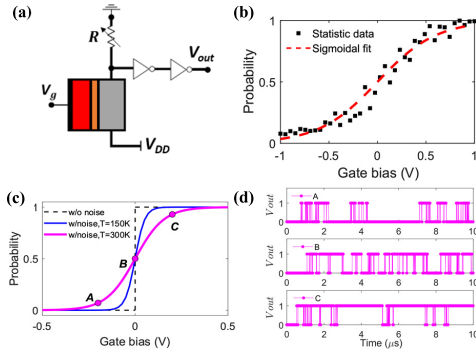


Fig. 4. (a) Proposed FE p-bit with external circuit setup. V_{DD} is 0.6 V, and R is adjustable resistor, ranging from 1 Ω to 100 k Ω . (b) Probabilistic curve of the FE p-bits, obtained from 10^4 sampled points. (c) P-curves with different stochasticity. A, B and C are three sampled points to illustrate the probability extraction. (d) Random bitstreams of sample gate bias points in (c) with total sampling point up to 10 μ s. V_{out} is normalized by $\frac{V_{DD}}{2}$.

the viscosity coefficients are assumed to follow the Gaussian distribution in a multidomain system, with original parameters as mean values and deviations σ up to 20% [25]. The I - V characteristics of the stochastic FeFET are shown in Fig. 3a by applying a series of pulses configured in the inset figure. The FeFET is first initialized by V_{init} , followed by a constant gate bias (V_{g1} , V_{g2} , ...) lasting for extensive time (~ 10 μ s). The I - V characteristics have finite broadening resulting from the current fluctuations (fluctuation range covered by blue lines in Fig. 3a) within the hold time of the pulse. The cyan dash lines are the forward/reverse sweep without the thermal noise. Under the relatively large gate bias (~ 3 V), the fluctuation range reduces as polarization reaches saturation, rendering minor capacitance and current fluctuations. A zoom-in view of the statistic distribution of current at each sample gate bias is shown in Fig. 3b, demonstrating the shifted current fluctuation range by the gate bias.

C. The Probabilistic Property of FE p-Bits

The FE p-bit design is shown in Fig. 4a, consisting of a single FeFET (width set as 100 nm), one adjustable resistor R , and two inverters. The design has significantly lower hardware cost comparing to the fully-CMOS p-bit design. The V_{out} forms a random bitstream with tunable probability by V_g . The adjustable resistor R and the inverters provide could binarize the analogue current output with the threshold voltage V_{th} . The probability curve (p-curve) could be extracted within the sampling time of 10 μ s (Fig. 4b). The probability at each gate bias is defined as the ratio of the sampled time when

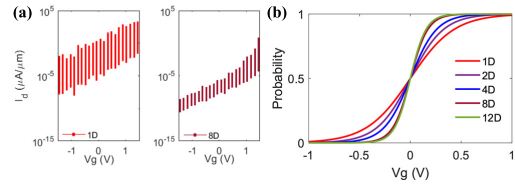


Fig. 5. (a) The I - V characteristics of FeFET consisting of a different number of domains (1D and 8D). (b) The p-curves extracted of FeFET with different number of domains from 1D to 12D.

V_{out} is $\frac{V_{DD}}{2}$ against the total sample time. Considering the dipole fluctuation time is at the scale of few ns, the probability is extracted over 10^4 samples. The statistical probability of V_{out} is in line with the sigmoidal relation, a key characteristic of the ideal p-bit. By excluding the effect of thermal noise, the p-curve of the deterministic I - V characteristics is a step-function (Fig. 4c). The intensity of dipole fluctuation, directly controlled by temperature in (3), leads to different slopes in p-curve (150 and 300 K). Therefore, the slope of the p-curve reflects the probabilistic behaviors of the p-bit. The extraction of the probability is further illustrated in Fig. 4d at the gate bias A, B, and C sampled in Fig. 4c. As the fluctuation range of the current could be shifted by the gate bias, there are different portions of time frames above the threshold in the sampled gate bias, leading to a different probability.

As the domain dynamic plays a crucial role in the FE system's stochasticity, its impact on p-bits' probabilistic behaviors needs to be evaluated. A series of stochastic FeFETs are simulated with a different number of domains in the FE system. The I - V characteristics of FeFET consisting of single and 8-domain (1D and 8D, respectively) are shown in Fig. 5a, demonstrating a significant reduction of the current fluctuation range as the number of domains increases. It could be attributed to the enhanced domain interaction and formation of domain walls with the increasing number of domains, leading to the narrowing of the current fluctuation range. The corresponding p-curves are also affected as they demonstrate different slopes (Fig. 5b). As the domain number changes from 1D to 12D, the p-curve gradually approaches that of a deterministic FeFET without thermal noise (Fig. 4c).

The various p-curves from the different number of domains suggest a wide range of probabilistic curves could be produced from the FE p-bit, providing extra design space through material engineering. The FE system with few domains is reported in multiple literatures [35], [36], providing some directions for the adjustment in FE domains. Besides, some circuit-level designs, such as the simulated annealing [5], could assist the adjustment of the effective probability of p-bits in circuit operations. Furthermore, the reliability issues are needed to be addressed as both FeFET and other circuit components in p-bit (Fig. 4a) could produce device-to-device variations under multiple factors, such as temperature and domain dynamics. Therefore, an FE p-bit design with optimal stochasticity needs to be co-optimized with other circuit components with reliability assessment, which is also depending on the different PC scenarios.

IV. CONCLUSION

A novel FE-based p-bit is proposed based on thermal noise-induced randomness and the domain dynamics are found to play the crucial role in p-bits' stochasticity. The proposed FE-based p-bits have the advantages of both ultralow hardware cost and scalability. This renders it a promising candidate for future large-scale PC applications.

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